

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING MANIT
BHOPAL

MINOR PROJECT

Medical Image Tampering Detection using Deep Learning in Deepfake Applications

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Problem Statement and Objectives

Problem Statement

Medical images can be manipulated with digital editing tools, potentially altering crucial diagnostic information. These subtle modifications are often hard to detect visually, which can result in erroneous medical decisions. Consequently, there is a pressing need for dependable systems to identify tampered medical images

Objectives

The goal of this project is to analyze medical images through image processing techniques to ascertain if they are authentic or altered. Additionally, it seeks to pinpoint the manipulated areas within the images and create a user-friendly interface for testing and analysis

Dataset Used – UCI Medical Image Tamper Detection Dataset

Overview

- The project uses the UCI Medical Image Tamper Detection Dataset.
- It contains medical images labeled based on authenticity and manipulation type.
- The dataset helps train the model to identify tampered and authentic image regions.



Class Labels

- True-Benign (TB) – no cancer
- True-Malignant (TM) – real cancer
- False-Benign (FB) – real cancer removed (tampered)
- False-Malignant (FM) – fake cancer added (tampered).

Dataset Preparation

- Images are organized into four folders based on the labels.
- The dataset is split into training, validation and testing sets in the ratio of 70, 15 & 15 respectively.
- All images are resized and normalized before training.
- Data is loaded using Pytorch DataLoader utilities for efficient processing.

Model Selection: Resnet50

Model Requirements

- Should detect very small tampered changes in medical images.
- Should extract deep and strong features like texture, edges, and abnormal patterns.
- Should support transfer learning because medical datasets are usually limited.
- Should train well even when the network is deep.
- Should work well for binary classification: tampered vs authentic.
- Should support explainability/localization like Grad-CAM heatmaps.
- Should be compatible with PyTorch/Torchvision and your preprocessing pipeline

Why ResNet50

- Deep architecture (50 layers) helps capture complex image features.
- Residual connections solve vanishing/degradation problems and make deep training easier.
- Works very well with pretrained ImageNet weights, so training becomes faster and more accurate on smaller datasets.
- Final layer can be easily changed for 2-class output in our project.
- Good balance of accuracy and computational cost.
- Supports Grad-CAM visualization, which is useful for highlighting suspicious regions in medical images.
- Easily integrates with our current medical image tampering detection pipeline.

ResNet50 was selected because it provides strong feature extraction, stable deep learning through skip connections, and effective classification of real and tampered medical images.

Model Architecture and Feature Extraction

ResNet50 is a deep convolutional neural network used for image classification. It uses residual learning with skip connections, which helps in training deep networks effectively and reduces the vanishing gradient problem. The model takes an input image of size $224 \times 224 \times 3$, then applies a 7×7 convolution layer and max pooling to extract basic features like edges and textures. Its main structure is made of Residual Bottleneck Blocks with 1×1 , 3×3 , and 1×1 convolution layers. Early layers extract low-level features, while deeper layers learn complex patterns and tampered regions. Finally, Global Average Pooling and a Fully Connected Layer classify the image into Real and Tampered classes.

Mathematical Equations Used

1. Residual Connection:

$$y = F(x) + x$$

Used to pass input directly to output, helping ResNet50 train deep networks effectively.

2. Convolution Operation:

$$y = W * x + b$$

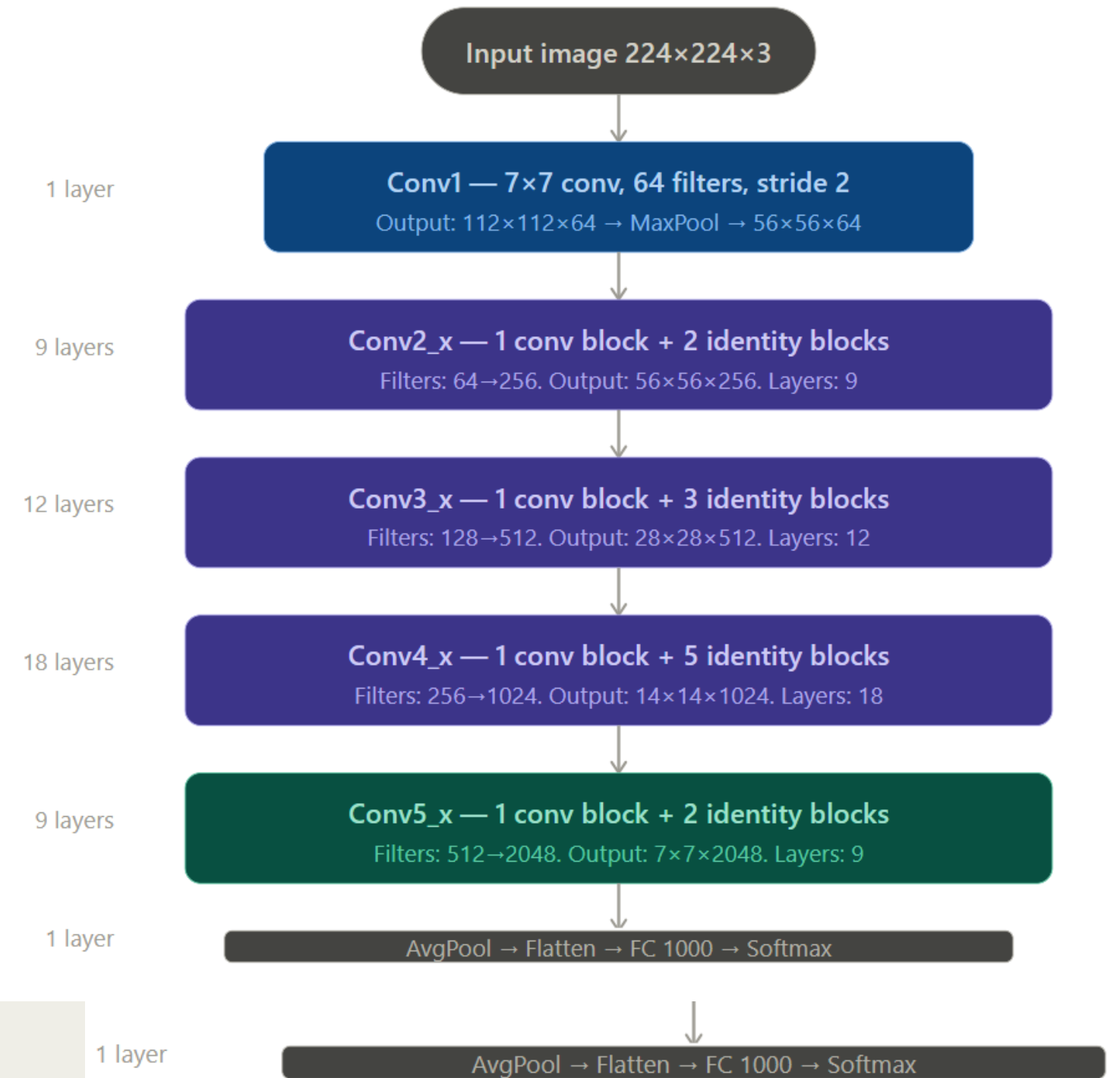
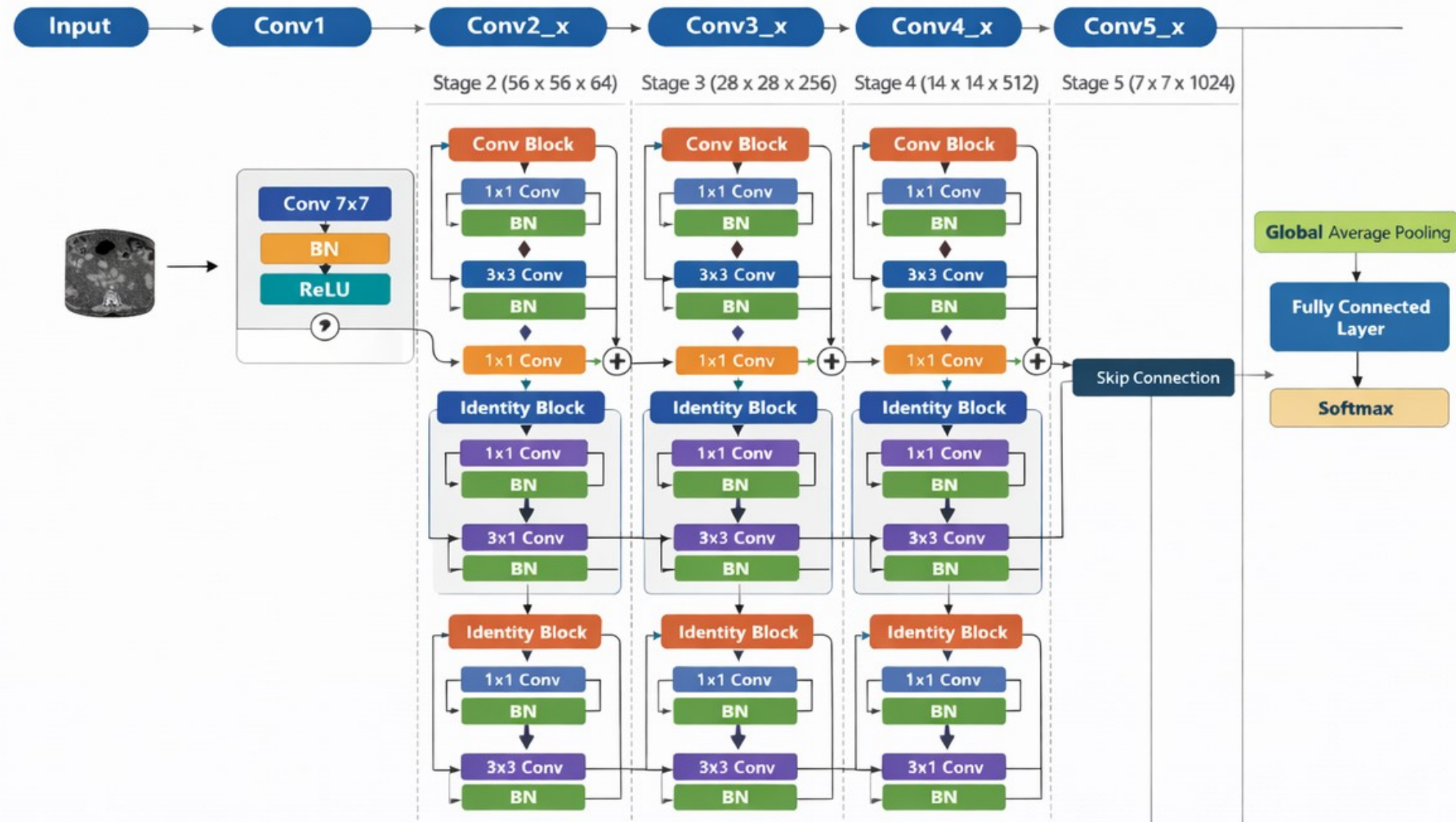
Used to extract important features such as edges, textures, and patterns from the image.

3. ReLU Activation Function:

$$f(x) = \max(0, x)$$

Used to introduce non-linearity so the model can learn complex patterns.

ResNet-50 Architecture



Preprocessing and Training Methodology

Preprocessing Pipeline:

Medical images in JPG and DICOM formats are first converted into a uniform form for analysis. Windowing is applied to DICOM images to enhance important anatomical details. All images are then resized to 224×224 , augmented to improve generalization, and normalized before being converted into PyTorch tensors for model input.

Model Training:

A pre-trained ResNet50 is used with transfer learning, where the final layer is modified for Real and Tampered classification. The dataset is split into 70:15:15 for training, validation, and testing. To handle imbalance, class weights and a weighted sampler are used. The model is trained for 15 epochs using the Adam optimizer and cross-entropy loss, while the best model is saved based on validation performance.

Mathematical Equations Used in Output:

1-Fully Connected Layer: $Z = Wf + b$ ----- Generate Output score from extracted features

2-SoftMax Function: $P(y_i) = \frac{e^{z_i}}{\sum e^{z_j}}$ ----- Output Score in class probabilities

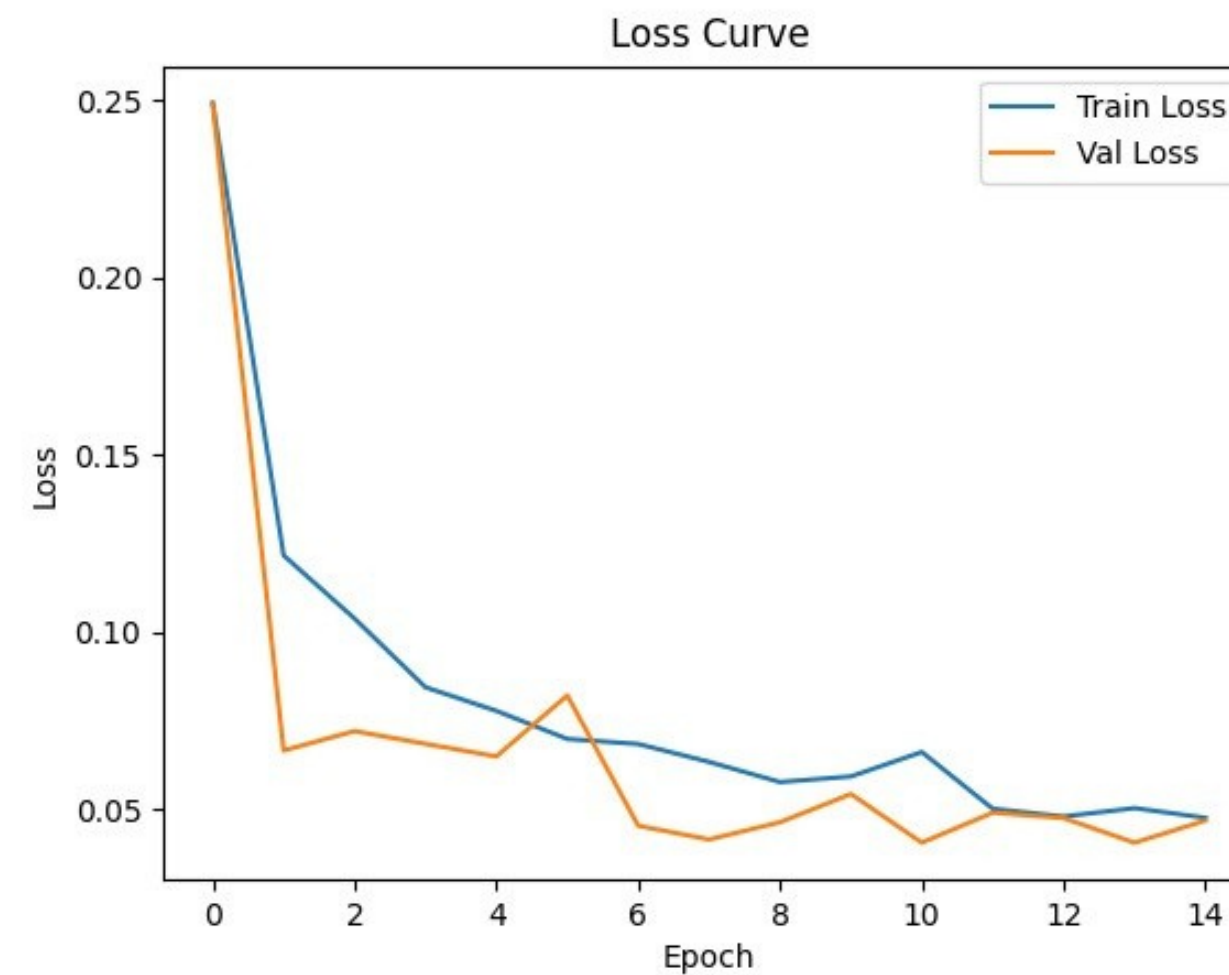
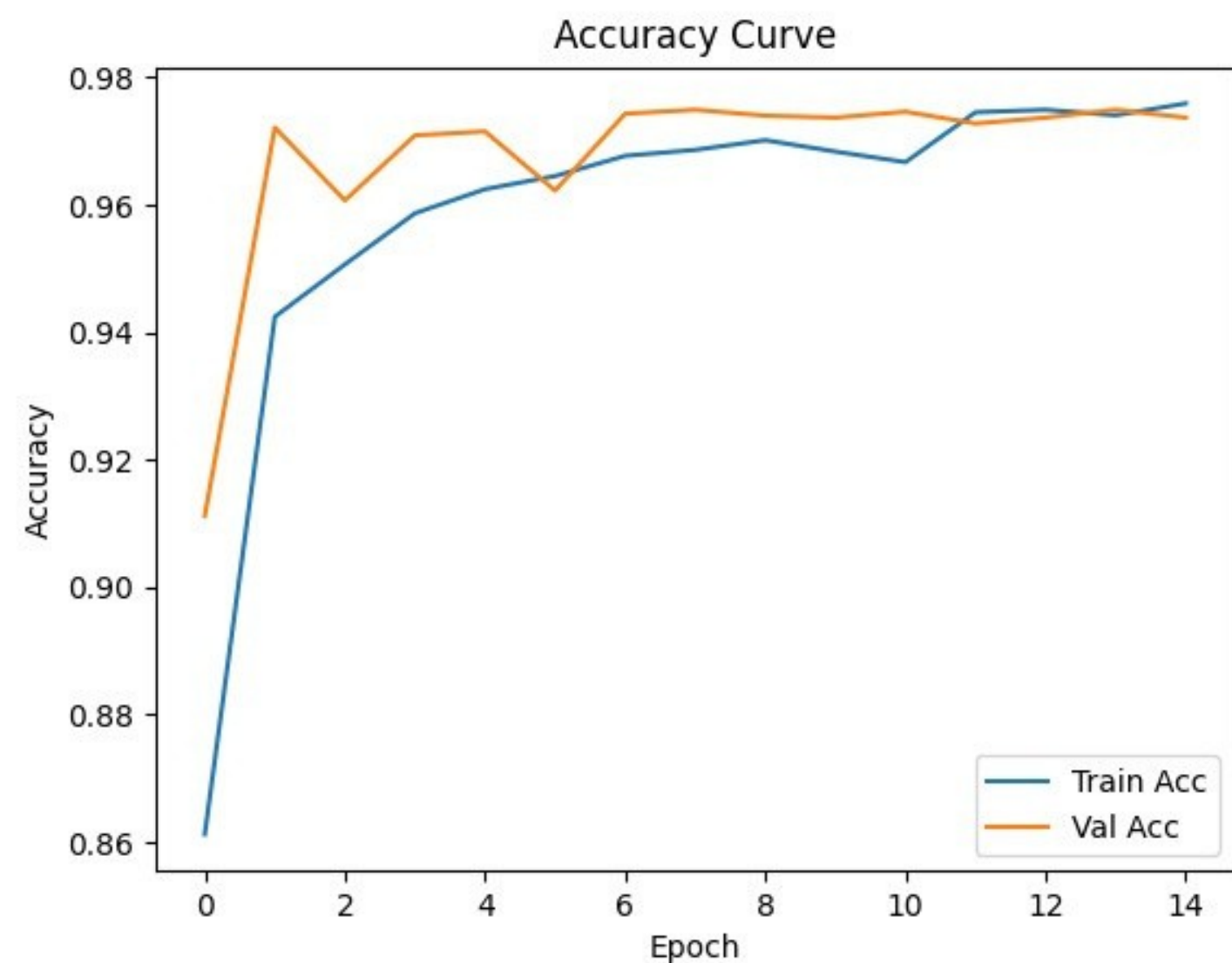
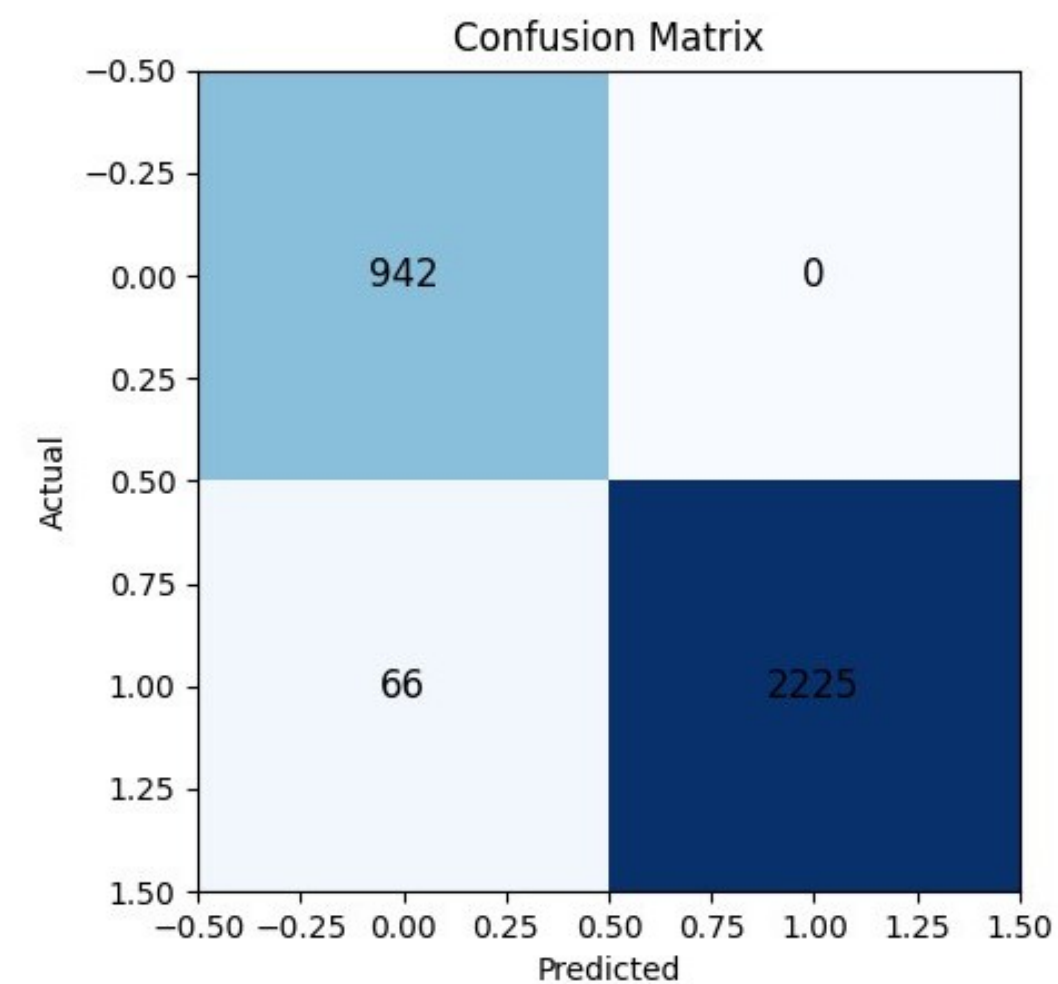
Result and Analysis

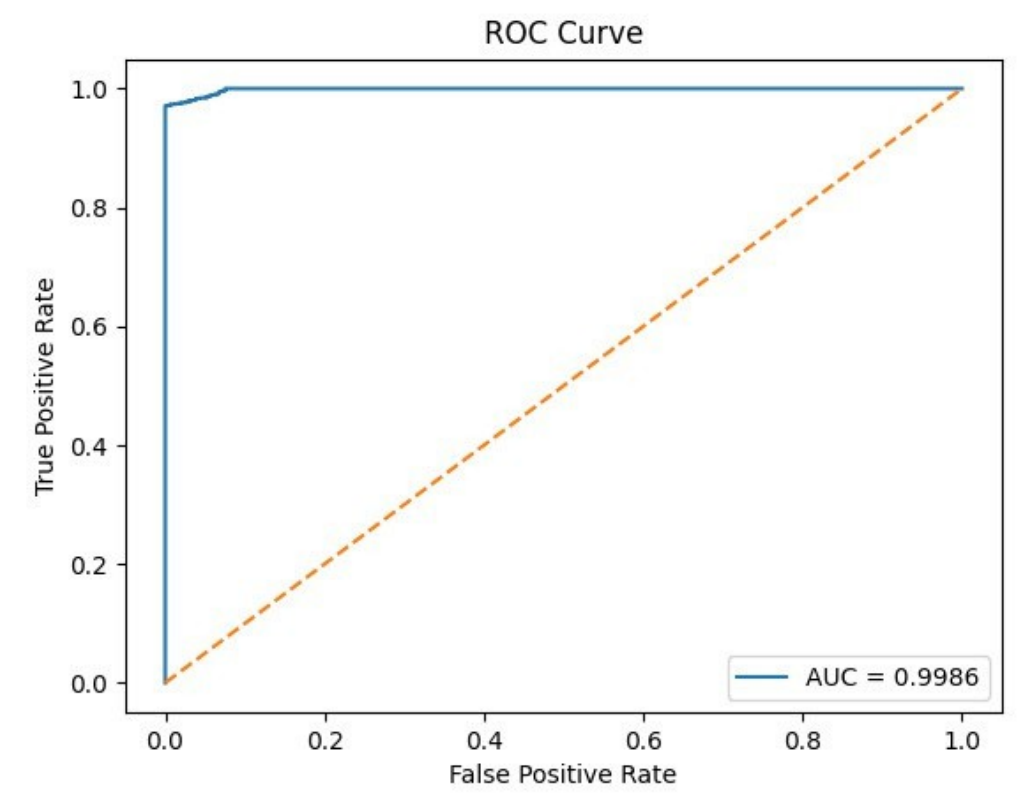
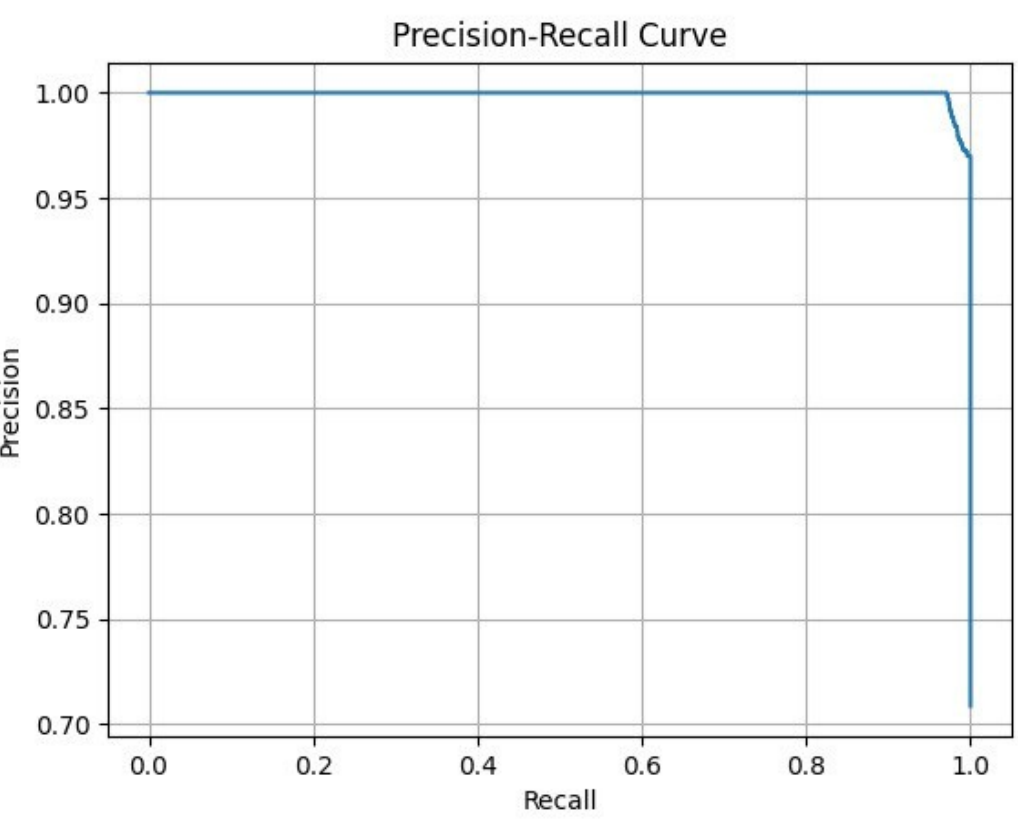
- The system successfully classifies the uploaded medical image as Authentic or Tampered.
- A probability score is displayed to show the confidence of the model prediction.
- For authentic images, the output shows normal classification results.
- For tampered images, the system also generates a heatmap to highlight the suspicious/tampered region.
- The heatmap improves interpretability by helping users visually identify the affected area.
- The average execution time for authentic images is about 0.45 s.
- The average execution time for tampered images is about 1.55 s due to additional heatmap processing.
- Overall, the system provides accurate, fast, and visually explainable results for medical image tampering detection.

Classification Report:

	precision	recall	f1-score	support
Authentic	0.94	0.99	0.97	942
Tampered	1.00	0.97	0.99	2291
accuracy			0.98	3233
macro avg	0.97	0.98	0.98	3233
weighted avg	0.98	0.98	0.98	3233

ROC-AUC: 0.9988995061446944





Analysis Results

Classification

Authentic

Confidence: 99.9%

Probabilities

Tampered: 0.1%

Authentic: 99.9%

Risk Level

0%

Analysis Results

Classification

Tampered

Confidence: 100.0%

Probabilities

Tampered: 100.0%

Authentic: 0.0%

Risk Level

100%

Tamper Localization (Heatmap)

A heatmap visualization of a bottle, showing a localized area of high intensity (red/yellow) at the base, indicating a tamper.